..... الرقم الجامعي : ..... الاسم :

Part I: 80 MCQ questions each of 1 point

Model

1. Software engineering is concerned with
   a. encouraging business unit to own modern applications
   b. enforcing organization to automate their business
   c. providing theories, methods and tools for professional software development
   d. introducing more and more job opportunities for software developers
2. Expenditure on software represents a significant fraction of GNP in all developed countries
   a. True b. False
3. Software costs more to develop than it does to maintain
   a. True b. False
4. According to ACM/IEEE Code of Ethics, software engineers ensure that their products and related modifications meet the highest professional standards possible
   a. True b. False
5. Increasing system complexity usually will lead to
   a. software project success c. more dependable software
   b. software project failure d. more and better services
6. Plan-driven processes are processes when
   a. some of the activities are planned c. all of the process activities are planned
   b. the project is of small or mid-size d. the system is not critical application
7. In practice, most practical processes include elements of both plan-driven and agile approaches
   a. True b. False
8. The main problems with Incremental development models are
   a. slow software delivery, systems structure degrading, and measuring project progress
   b. measuring project progress and possible systems structure degrading
   c. getting customer feedback, slow software delivery, and system structure degrading
   d. measuring project progress, getting customer feedback, slow software delivery
9. Integration and configuration models depend on building software from scratch
   a. True b. False
10. Requirements validation is used to
    a. define the internal structure of system components
    b. present what do the system stakeholders require or expect from the system
    c. define the requirements in detail
    d. check the validity of the requirements
11. Programming is an individual activity with no standard process
    a. True b. False
12. Incremental delivery usually complies with procurement model of many organizations, where the complete system specification is part of the system development contract
    a. True b. False
13. Verification and validation (V & V) is intended to show that a system conforms to its specification and meets the requirements of the system customer
    a. True b. False
14. System prototyping and Incremental delivery could cope with changing requirements
    a. True b. False
15. Approachesto processimprovement can include
    a. simplifying code designing c. following the process maturity approach
    b. documenting testing process d. using incremental delivery

16. In Agile development methods
    a. Team usually depends on native programming languages and commonly used utilities
    b. Team depends on reusable codes and does not build any new codes
    c. Team builds their own codes and do not reuse any available components
    d. Team uses extensive tools to support software development
17. Agile manifesto states:
    a. We have come to value "following a plan" over "responding to change"
    b. We have come to equally value both "responding to change" and "following a plan"
    c. We have come to value "responding to change"over "following a plan"
    d. We have come to value "responding to change" but not "following a plan"
18. Extreme Programming is
    a. one of the most well-known Plan driven models
    b. one of the most well-known Agile methods
    c. one of the versions of Waterfall model
    d. one of the common use programming techniques
19. Main extreme programming practices are
    a. Incremental planning, Small releases, Simple design, Test-first development
    b. Incremental planning, Complete releases, Complex design, Test-first development
    c. Incremental planning, Small releases, Complex design, Test-first development
    d. Plan-driven system, Small releases, Simple design, Test-first development
20. In Test-driven development
    a. Tests are designed and executed after implementation
    b. Tests are run after coding using a set test data
    c. Tests are run manually
    d. Tests are written as programs and designed before code
21. Product backlog is a list of 'to do' items which the Scrum team must tackle
    a. True b. False
22. While Scaling up agile methods to large systems
    a. more focusing on co-located team communication is needed
    b. cross-team communication mechanisms have to designed and used
    c. focus on team communication is done as it is applied in small systems
    d. focus on co-located team communication is an essential customer requirement
23. Project managers in large organizations usually use agile project management approaches
    a. True b. False
24. User requirements are statements in natural language plus diagrams of the services that the system provides
    a. True b. False
25. Functional user requirements should describe the details of the required services
    a. True b. False
26. Functional system requirements may be high-level statements of what the system should do
    a. True b. False
27. Specifying a particular IDE, programming language, or development technique is
    a. a type of functional requirements c. a type of non-functional requirements
    b. a type of user functional requirements d. a type of system functional requirements
28. Usability, Efficiency, Dependability, and Security are considered as
    a. External non-functional requirements c. Organizational non-functional requirements
    b. Product non-functional requirements d. Customer functional requirements
29. Metrics for specifying the non-functional requirement "Reliability" could be
    a. Mean time to failure, Rate of failure occurrence, Availability
    b. Event response time, Rate of failure occurrence, Availability
    c. Event response time, Probability of unavailability, Usability
    d. Event response time, Rate of failure occurrence, Usability

30. Metrics for specifying the non-functional requirement "Ease of use" could be
   a. Mean time to failure, Rate of failure occurrence, Availability
   b. Training time, Number of help functions, Availability
   c. Training time, Number of help frames, Number of user clicks/function
   d. Event response time, Number of help functions, Usability

31. Ethnography is a technique used in
   a. Requirements validation
   b. Requirements discovery
   c. Requirements specification
   d. Requirements classification

32. Writing requirements specifications using Structured specifications technique
   a. works well for all types of business and embedded control system requirements
   b. works well for only some business system requirements
   c. does not work for all types of embedded control systems requirements
   d. works well for embedded control systems requirements

33. Requirements validation techniques include
   a. Requirements compatibility, Test-case generation, or Interviews
   b. Requirements reviews, Prototyping, or Interviews
   c. Requirements reviews, Interviews, or Test-case generation
   d. Requirements Test-case generation, Prototyping, or Interviews

34. Requirements Verifiability means
   a. Are the requirements realistically testable?
   b. Are the requirements fulfilling what the customers need?
   c. Are the requirements covering all customers' needs?
   d. Are the requirements realistically applicable?

35. In requirements change management
   a. change will be implemented directly when it is detected
   b. change should be analysed and assessed for implementation
   c. change will be ignored as it will be detected while testing phase
   d. change will be managed as new independent requirement

36. Modeling user-system interaction and system-to-system interaction could be represented by
   a. Activity diagram and Sequence
   b. Use case and State diagram
   c. Use case and Sequence diagram
   d. Class diagram and Sequence diagram

37. Structural models that show the statics of the system could be represented by using
   a. Use case diagrams
   b. Use case and Sequence diagrams
   c. Activity and State diagrams
   d. Class diagrams

38. The dynamic behavior of a data-driven-system could be properly modeled using
   a. State diagram
   b. Class diagram and Sequence diagram
   c. Use case diagram and Activity diagram
   d. Activity diagram and Sequence diagram

39. In Model-driven architecture the generated models are very suitable for implementation
   a. True
   b. False

40. The right sequence of models used to build a system using Model-driven engineering is
   a. Platform independent model, Computational independent model, then Platform specific model
   b. Platform independent model, Platform specific model, then Computational independent model
   c. Computational independent model, Platform independent model, then Platform specific model
   d. Platform specific model, Computational independent model, then Platform independent model

41. Companies prefer using MDA for the long-lifetime systems
   a. True
   b. False

42. When the "Performance" is a main system characteristic
   a. It is recommended to use layered architecture
   b. It is recommended to localize critical operations and minimize communications
   c. It is recommended to use a distributed system with small components
   d. It is recommended to use redundant components with short codes

43. When the "Safety" is a main system characteristic
    a. It is recommended to use layered architecture with critical assets in inner layers
    b. It is recommended to use a distributed system with small components
    c. It is recommended to use redundant components with short codes
    d. Localize safety-critical features in a small number of sub-systems
44. Architectural pattern should include information about
    a. when it can be used and how to realize it
    b. when it can be used and when it will not be useful
    c. how it can be built and implemented
    d. how it can be used and tested
45. Using MVC architectural pattern
    a. will avoid involving additional code when the data model is simple
    b. does not allow the data to change independently of its representation and vice versa
    c. supports presentation of the same data in different ways
    d. prevents introducing additional functionality when the user interaction is simple
46. Layered architectural pattern is used when
    a. there are multiple ways to view and interact with data
    b. there are multiple processes sharing data sources
    c. there is a need to build new facilities on top of existing systems
    d. there are multiple users accessing data sources over the network
47. The architectural pattern that supports efficient data sharing mechanism is
    a. Repository architecture
    b. Model-View-Controller architecture
    c. Layered architecture
    d. Pipe and Filter architecture
48. Client-server architectural pattern is used when
    a. there are multiple ways to view and interact with data
    b. data-driven system push data in shared source to trigger an action
    c. there is a need to build new facilities on top of existing systems
    d. there are a set of stand-alone servers which provide specific services
49. Pipe and filter architectural pattern is usually used when components use heterogeneous data structure
    a. True
    b. False
50. The Observer model is used when
    a. multiple objects have the same display
    b. multiple users view multiple objects in the system
    c. multiple displays of the same state are required
    d. single view is required for multiple objects
51. At System reuse level, we reuse objects from a library rather than writing the code
    a. True
    b. False
52. Which of the following is a cost factor in reuse software
    a. the cost of redesigning and modifying the reused software
    b. the cost of building system models for the reuse software
    c. the cost of coding and documenting the reuse software
    d. the cost of integrating the reusable software elements with each other
53. Configuration management activities are
    a. component selection, adaptation, and system integration
    b. subsystem adaptation, integration, and documentation
    c. version management, system integration, and problem tracking
    d. system design, integration, and testing
54. Development platform usually is the same as the execution platform
    a. True
    b. False

55. The business model in Open-source development is reliant on
   a. selling a software product but not selling the support for that product
   b. selling a software product and selling its documentation
   c. selling the support for the product but not selling the software product
   d. selling a software product and selling support for that product
56. Developers use open-source software in building new products under GPL license
   a. may make those new products open source
   b. must make those new products open source
   c. should pay for the open-source license if they want to sell their products
   d. should not make those new products open source
57. The Berkley Standard Distribution (BSD) license is
   a. a reciprocal license
   b. not used in producing open-source code
   c. a non-reciprocal license
   d. used only in producing open-source code
58. Testing can reveal the presence of errors and ensure their absence
   a. True b. False
59. For generic software, Program Testing demonstrates to the developer and the customer that
   a. the software product meets its requirements
   b. the software product is designed and implemented according to standard approaches
   c. all customer requirements had been realized
   d. the software release is ready for user testing
60. V & V confidence in software testing is independent of Software purpose
   a. True b. False
61. Software inspections could test incomplete versions of the software
   a. True b. False
62. Software inspections are very limited to code defects
   a. True b. False
63. Unit testing should focus on
   a. testing component interfaces
   b. testing all separate components and their interactions
   c. testing component interactions
   d. testing functionality of objects or methods of a component
64. Unit testing using values that avoid the boundaries of input sequence is a recommended guideline
   a. True b. False
65. Use-case and sequence diagrams can be used in
   a. System testing c. Component testing
   b. Unit testing d. both Unit and Component testing
66. In Regression testing, the new program is only tested using all previous test data
   a. True b. False
67. Release testing is intended to test the system for use outside of the development team
   a. True b. False
68. User Testing can include these types of testing
   a. Alpha testing, Unit testing, Release testing
   b. Unit testing, Component testing, and Beta testing
   c. Alpha testing, Beta testing, and Acceptance testing
   d. White-box testing, System testing, and Release testing
69. The first stage of change implementation in software evolution may need
   a. redesign the changed components
   b. program understanding when the original developers are not available
   c. code refactoring
   d. requirements testing

70. In software evolution, we do not have a handover problem if
   a. the original developers followed agile, but the evolution team followed plan-based
   b. the original developers followed plan-based, but the evolution team followed agile
   c. the original developers and the evolution team followed plan-based
   d. the original and evolution developers followed different but well-defined methods

71. The strategy chosen on legacy system management depends on
   a. the type of change
   b. the system quality and business value
   c. business requirements and customers' needs
   d. the type of business

72. Legacy system of "Low-quality, high-business value" is recommended to
   a. be scrapped
   b. be replaced with a new custom system
   c. be re-engineered or replaced
   d. continue in operation using normal maintenance

73. The quality level of legacy systems depends on how the business process is effectively supported by the legacy application software
   a. True
   b. False

74. The business value of legacy systems depends on the type of used process model
   a. True
   b. False

75. Factors used in application assessment to predict the quality level of legacy system include
   a. Test data availability, Data model, Documentation, and Support cost
   b. Test data availability, Data model, Support cost, and Interoperability
   c. Test data availability, Data model, Documentation, and Programming language
   d. Data model, Documentation, Application Performance, and Interoperability

76. Changing a system so that it operates in a different OS environment
   a. Fault repairs maintenance
   b. Functionality addition
   c. Functionality modification
   d. Environmental adaptation

77. As Maintenance is more frequent, the system structure will
   a. degrade
   b. upgrade
   c. change more rapidly
   d. not be changed

78. Software re-engineering is
   a. rebuilding the functions of the system
   b. restructuring part of the system and changing its functionality
   c. restructuring part of the system without changing its functionality
   d. changing system functionality without restructuring part of the system

79. Specifications in Generic software products are made by
   a. customer
   b. software stakeholders
   c. owner's managers
   d. software developer

80. The cost of re-engineering depends on the extent of the data conversion which is required
   a. True
   b. False

1. [10 points] Based on your experience with a bank ATM, draw an activity diagram that models the data processing involved when a customer withdraws cash from the machine.

2. [10 points] Draw the state diagram of the control software for the camera on your mobile phone. Ignore the flash if you have one on your phone.